Limiting Betti-number distributions in preferential-attachment clique complexes

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Abstract

We study the finite clique complexes of sequential affine preferential-attachment graphs. For every $q \geq 2$, $m \geq 2q$ and $\delta > -m$ with $x = (m+\delta)/(2m+\delta) < 1/(2q)$, the qth rational Betti number, divided by its expectation, converges to a nondegenerate random variable. We identify its law as a normalized sum of products of attachment weights over induced octahedral subgraphs in an explicit independent-Beta construction. The limit is positive almost surely, has zero essential infimum and has every positive moment. In particular, its upper tail is not regularly varying with any finite positive exponent. The proof compares all octahedral link contributions with finite-time homology; larger link cores have lower order, and an earlier cone makes each fixed base contribute to subsequent births.

1 Introduction

Preferential attachment produces growing graphs in which vertices of large degree attract more edges. Their clique complexes record higher-order connections: a complete subgraph is filled by a simplex, and the Betti numbers measure the homology that remains. The variability of these numbers across independently generated graphs is a different question from their average growth.

In the affine model, each new vertex adds m edges and the attraction of an old vertex is its current degree plus δ. Write $x = (m + \delta)/(2m + \delta)$. For the sequential version considered here, Siu, Samorodnitsky, Yu and He [4] proved that the expected qth Betti number has order T^{1-2qx} when $q \geq 2$, $m \geq 2q$ and $x < 1/(2q)$. Siu [3] then obtained matching upper and lower bounds in probability, up to any diverging multiplicative factor. The latter proof counts common neighbors of a fixed induced octahedral sphere and controls lower-order killing terms. These bounds locate the scale but leave the limiting distribution unidentified. In Section 1.2 of the author preprint [3], mean-normalized convergence and a power-law tail were suggested by numerical simulations.

We prove the convergence throughout this strict polynomial-growth regime. The limit is an explicit random series over all induced octahedral bases, and it has every positive moment. This gives a nonconstant limiting law while refuting regular variation with a finite positive exponent. The normalization is the ensemble expectation of the Betti number, as in the distributional question; no empirical normalization is involved.

Two issues separate a distributional result from the existing order bounds. The count must include all base spheres, with a summable bound on their large labels. It must also track whether their link homology actually produces new homology in the finite complex. We use the independent-Beta description of the attachment process to control the first issue. For the second, deleting vertices of small degree reduces a typical link to a single octahedral sphere. Every fixed base eventually acquires an earlier cone, after which its one-dimensional

link homology maps to zero in the old complex and contributes one birth. The remaining link cores require an additional neighbor and have lower order.

Random environments also describe non-Gaussian triangle-count limits in the same attachment model, as shown recently by Dey and Terlov [1]; their triangle observable counts edge multiplicity. Here the coefficient additionally retains the induced graph on the base vertices. Joint weight and edge estimates handle that dependence, including overlapping bases in every fixed moment. A conditional estimate obtained by taking one Beta variable close to one shows that the coefficient can be arbitrarily small with positive probability, proving nondegeneracy.

Section 2 states the exact model and the limiting law. Section 3 establishes the weight and counting estimates. Section 4 compares the counts with finite homology, and Section 5 proves nondegeneracy and the tail conclusion. Homology is over $\mathbb{Q}$ throughout. Constants may depend on the fixed model parameters and, where indicated, on a fixed moment order; they do not depend on time or vertex labels.

2 The model and the limiting law

Fix integers $q \geq 2$, $m \geq 2q$ and a real number $\delta > -m$ such that

$$a = m+\delta > 0, \qquad c = 2m+\delta = m+a, \qquad x = \frac{a}{c} < \frac{1}{2q}, \qquad k = 2q, \qquad \rho = 1-kx > 0. \tag{1}$$

The graph G_2 consists of two vertices joined by m parallel edges. At each time $t \geq 3$, vertex t adds m edges, one at a time, to old vertices. After the first $b-1$ edges, the next endpoint $v < t$ is chosen with probability

$$\frac{d_{t,b-1}(v) + \delta}{2m(t-2) + (t-1)\delta + (b-1)}, \qquad 1 \leq b \leq m. \tag{2}$$

Degrees count multiplicity. There are no self-loops. Write X_T for the clique complex of the underlying simple graph, and $\beta_q(T) = \dim_{\mathbb{Q}} H_q(X_T; \mathbb{Q})$.

The following construction puts all times on one probability space. Let $\psi_1 = 1$ and let

$$\psi_j \sim \mathrm{Beta}(a, b_j), \qquad b_j = c(j-1) - m, \qquad j \geq 2, \tag{3}$$

be independent; $b_2 = a$ and all b_j are positive. Given (ψ_j), every edge from t chooses its old endpoint independently with probabilities

$$p(v,t) = \psi_v \prod_{v<j<t} (1-\psi_j), \qquad v < t. \tag{4}$$

Their sum is one by telescoping. Independent uniform variables on $(0,1)$ can be used to make these choices. At $t = 2$, all endpoints are 1. Denote this infinite graph by G and use its simple graph when speaking of induced subgraphs.

Lemma 2.1. *The restrictions of this construction have exactly the joint sequential graph law in* (2).

Proof. This is the Pólya representation used in [3, Definition 5.1 and Theorem 5.2 of the author preprint]. We include the verification to fix the precise variant. At the completion of the birth of vertex j, the total weight of the vertices before j is

$$2m(j-2) + (j-1)\delta + m = c(j-1) - m = b_j,$$

and vertex j has weight a. A later endpoint at j contributes ψ_j to the likelihood. An endpoint before j from a later vertex contributes $1-\psi_j$; other endpoints do not change its factor.

Conditional on any finite endpoint history, the variables belonging to eligible vertices therefore remain independent, with posterior law $\mathrm{Beta}(w_j, W_{j-1})$, where w_j is the current weight of j and $W_i = \sum_{v \le i} w_v$. This holds during a partial new birth as well. Taking the posterior expectation in (4) yields

$$\frac{w_v}{W_v} \prod_{v<j<t} \frac{W_{j-1}}{W_j} = \frac{w_v}{W_{t-1}}.$$

The denominator after $b-1$ new edges is precisely the one in (2). Initially $w_1 = w_2 = a$ and $\psi_2 \sim \mathrm{Beta}(a, a)$, while the m initial edges are deterministic. This proves the claim for every finite ordered history. □

We will show that the following limits exist and are positive almost surely:

$$R_v = \lim_{n\to\infty} n^x \prod_{v<j\le n} (1-\psi_j), \qquad D_v = \psi_v R_v. \tag{5}$$

They describe the fixed-vertex attachment probabilities: $t^x p(v,t) \to D_v$. Let $\mathcal{O}_q$ be the graph K_{2q} with a perfect matching removed. Its clique complex is the join of q two-point spaces, an octahedral $(q-1)$-sphere. For every k-element set $S \subset \mathbb{N}$, put

$$I_S = \mathbf{1}\{G[S] \cong \mathcal{O}_q\}, \qquad A = \sum_{\substack{S\subset\mathbb{N}\\ |S|=k}} I_S \prod_{v\in S} D_v. \tag{6}$$

Here $G[S]$ is the induced simple graph, so I_S imposes the missing matching as well as the present edges. The series is initially understood as a nonnegative extended random variable; its finiteness is part of the result.

Theorem 2.2. *For every parameter triple in (1), the weights in (5) are positive and finite almost surely, and*

$$A > 0 \text{ almost surely}, \qquad \operatorname{ess\,inf} A = 0, \qquad 0 < \mathbb{E}A < \infty, \qquad \mathbb{E}A^p < \infty \quad (p > 0).$$

In the construction (3)–(4),

$$\frac{\beta_q(T)}{T^\rho} \xrightarrow{L^1} \frac{m^{\underline{k}}}{\rho} A, \qquad \frac{\mathbb{E}\beta_q(T)}{T^\rho} \longrightarrow \frac{m^{\underline{k}}}{\rho}\mathbb{E}A, \tag{7}$$

where $m^{\underline{k}} = m(m-1)\cdots(m-k+1)$. Consequently the mean-normalized Betti number converges in distribution to

$$W_{q,m,\delta} = \frac{A}{\mathbb{E}A}, \tag{8}$$

a nondegenerate random variable with mean one. Its upper tail does not satisfy

$$\lim_{z\to\infty} \frac{\mathbb{P}(W_{q,m,\delta} > uz)}{\mathbb{P}(W_{q,m,\delta} > z)} = u^{-\alpha}, \qquad u > 0, \tag{9}$$

for any finite $\alpha > 0$.

The law in (8) is specified by independent Beta variables, independent endpoint uniforms and the convergent series (6). All parameters are fixed before $T \to \infty$.

3 Attachment weights and octahedral counts

3.1 Joint moments

We first obtain bounds uniform in all vertex labels. For a directed simple graph F whose edges point from larger to smaller labels, let $d_-(v)$ and $d_+(v)$ be its incoming and outgoing degrees, respectively. The event $F \subset G$ requires its edges and places no restriction on other edges. The number of vertices and edges in this lemma is fixed.

Lemma 3.1. *All the limits in (5) exist and are positive almost surely. For nonnegative integers r_v on a finite set V containing the vertices of F,*

$$\mathbb{E}\left[\mathbf{1}\{F \subset G\} \prod_{v \in V} D_v^{r_v}\right] \le C \prod_{v \in V} v^{-(1-x)(r_v + d_-(v)) - x d_+(v)}. \tag{10}$$

The constant depends only on m, δ, the finite size bounds and $\sum_v r_v$, not the labels. Extend $r_v, d_\pm(v)$ by zero outside V. For $K \ge 2$, define

$$\ell_K = r_K + d_-(K), \qquad h_K = \sum_{v<K} r_v + \#\{u \to v \in F : v < K < u\}.$$

The conditional version of (10), with $\psi_K = z \in (0,1)$, holds with the additional factor $CK^{\ell_K}(1-z)^{h_K}$, uniformly in K and z.

Proof. Write $Y_j = -\log(1-\psi_j)$. For $0 \le u < 1$,

$$0 \le -\log(1-u) - u \le \frac{u^2}{1-u}, \qquad [-\log(1-u)]^2 \le \frac{u^2}{(1-u)^2}.$$

For all sufficiently large j, beta integrals give

$$\begin{aligned} \mathbb{E}\psi_j &= \frac{a}{cj - 2m} = \frac{x}{j} + O(j^{-2}), \\ \mathbb{E}\frac{\psi_j^2}{1-\psi_j} &= \frac{a(a+1)}{(a+b_j)(b_j-1)} = O(j^{-2}), \\ \mathbb{E}\frac{\psi_j^2}{(1-\psi_j)^2} &= \frac{a(a+1)}{(b_j-1)(b_j-2)} = O(j^{-2}). \end{aligned}$$

Thus $\mathbb{E}Y_j = x/j + O(j^{-2})$ and $\mathrm{Var}(Y_j) = O(j^{-2})$. The finitely many early indices have finite squared logarithmic moments, since $b_j > 0$. The independent series criterion [2, Theorem 2.5.6] shows that $\sum_{j \ge 2}(Y_j - \mathbb{E}Y_j)$ converges almost surely. The summable expectation error and the convergence of the harmonic sum minus $\log n$ now prove (5). A countable intersection handles every v.

Given the environment, assign distinct edge slots to the required older neighbors of each vertex of F. A union bound gives

$$\mathbb{P}(F \subset G \mid (\psi_j)) \le m^{|E(F)|} \prod_{u \to v \in F} p(v,u). \tag{11}$$

If a required outdegree exceeds m, the event is empty. To estimate the right side multiplied by the weight product, first use $D_{v,n} = \psi_v n^x \prod_{v<j\le n}(1-\psi_j)$, with n larger than all labels. Independence reduces the expectation to beta factors

$$\mathbb{E}[\psi_j^{\ell_j}(1-\psi_j)^{h_j}] = \frac{(a)^{\overline{\ell_j}}(b_j)^{\overline{h_j}}}{(a+b_j)^{\overline{\ell_j + h_j}}}, \tag{12}$$

where a rising factorial is indicated by an overline, and

$$\ell_j = r_j + d_-(j), \qquad h_j = \sum_{v<j} r_v + \#\{u \to v \in F : v < j < u\}.$$

Both exponents have a fixed bound. At the finitely many sites with $\ell_j > 0$, the factor is at most $Cj^{-\ell_j}\exp(-xh_j/j)$. At other sites its logarithm is $-xh_j/j + O(j^{-2})$. These statements follow by taking logarithms of the finite products in (12); the constants also cover the finitely many small indices. Consequently the complete product is bounded by

$$Cn^{x\sum_v r_v}\prod_v v^{-r_v-d_-(v)}\exp\left(-x\sum_{j=2}^{n}\frac{h_j}{j}\right).$$

For each weight, the harmonic sum is $\log(n/v) + O(1/v)$, and for an edge $u \to v$ it is $\log(u/v) + O(1/v)$. Substitution gives

$$Cn^{x\sum_v r_v}\prod_v v^{-r_v-d_-(v)}\prod_v (n/v)^{-xr_v}\prod_{u\to v}(u/v)^{-x},$$

which is the product in (10). Vertex 1 has $\psi_1 = 1$ and label 1, so it introduces no beta factor or label power. Fatou's lemma passes to the limiting weights.

For the conditional assertion take $n \geq K$ as well. Only the Kth factor changes, to $z^{\ell_K}(1-z)^{h_K}$. The beta integral (12) has the uniform lower bound $C^{-1}K^{-\ell_K}$ at that site, for fixed exponent bounds. Its replacement therefore costs at most $CK^{\ell_K}(1-z)^{h_K}$. Fatou's lemma applies with z fixed, proving the stated conditional estimate. □

3.2 Summing the labels

The following elementary observation is what makes early bases dominate. It will also control unions of overlapping bases.

Lemma 3.2. *Let F be an ordered simple graph with minimum degree at least $d \geq 2$, and let $r_v \geq 1$ be integers. Put*

$$\gamma_v = -(1-x)(r_v + d_-(v)) - xd_+(v).$$

If $0 < x < 1/2$, the sum of $\prod_v v^{\gamma_v}$ over all increasing positive vertex labels is finite. For any fixed ordered exponent sequence $\gamma_1, \dots, \gamma_h$ obtained this way,

$$\sum_{K<v_1<\cdots<v_h}\prod_i v_i^{\gamma_i} \leq CK^{h+\sum_i\gamma_i}, \tag{13}$$

$$\sum_{K=v_1<v_2<\cdots<v_h}\prod_i v_i^{\gamma_i} \leq CK^{h-1+\sum_i\gamma_i}. \tag{14}$$

Proof. For a nonempty terminal segment U of the ordered vertices, all edges crossing its boundary point toward earlier vertices outside U. Hence

$$\begin{aligned}\sum_{v\in U}(\gamma_v+1) &= |U| - (1-x)\sum_{v\in U} r_v - |E(U)| - x|E(U,U^c)| \\ &\leq x|U| - |E(U)| - x|E(U,U^c)| \\ &\leq -(d-1)x|U| - (1-2x)|E(U)| < 0.\end{aligned} \tag{15}$$

The last step uses $2|E(U)| + |E(U,U^c)| \geq d|U|$. Place v_i in the dyadic interval $[2^{n_i}, 2^{n_i+1})$. The sum in those intervals is at most a constant times $2^{\sum_i(\gamma_i+1)n_i}$, with $n_1 \leq \cdots \leq n_h$. Writing the n_i as successive nonnegative differences turns the full sum into a product of geometric sums,

whose exponents are precisely the negative suffix sums (15). For (13), start the dyadic intervals at K; their initial factor is $K^{h+\sum_i \gamma_i}$. For (14), fix the first label and sum the remaining $h-1$ labels in the same way. All their suffix sums are still negative, and the initial factor is $K^{h-1+\sum_i \gamma_i}$. □

For an octahedral graph F on ordered labels $v_1 < \cdots < v_k$, write

$$f_F(v_1, \ldots, v_k) = \prod_{i=1}^{k} v_i^{-(1-x)(1+d_-(v_i))-xd_+(v_i)}. \tag{16}$$

There are finitely many perfect matchings on k ordered positions. Lemma 3.1 and Lemma 3.2, with $d = k-2 \geq 2$, give a summable bound on $\mathbb{E}[I_S \prod_{v\in S} D_v]$. In particular $\mathbb{E}A < \infty$.

Proposition 3.3. *The series A has every positive moment.*

Proof. For an integer $p \geq 1$, expand the pth power of finite partial sums of (6). Classify the p bases by their overlap pattern and by their missing matchings. There are finitely many such patterns at fixed p and k. In each pattern, the union of the required edges forms a graph F of minimum degree at least $k-2$, because every vertex belongs to an octahedral base. Its weight multiplicity r_v is the number of bases containing it, so $1 \leq r_v \leq p$. Ignoring extra nonedge requirements only increases the probability; an inconsistent pattern or a required outdegree larger than m contributes zero. Otherwise (10) bounds the weighted expectation, and Lemma 3.2 sums its labels. This bound is finite for every pattern. Monotone convergence proves $\mathbb{E}A^p < \infty$. For an arbitrary real $p > 0$, use $A^p \leq 1 + A^{\lceil p \rceil}$. □

3.3 Common neighbors and the complete count

For a fixed k-set S and $t > \max S$, let $Y_{S,t}$ indicate that t is joined to every vertex of S. Put $N_S(T) = \sum_{\max S < t \leq T} Y_{S,t}$.

Lemma 3.4. *For every fixed S,*

$$\frac{N_S(T)}{T^\rho} \longrightarrow \frac{m^{\underline{k}}}{\rho} \prod_{v\in S} D_v \quad \textit{almost surely and in } L^1. \tag{17}$$

Moreover $\sup_T \mathbb{E}[(N_S(T)/T^\rho)^2] < \infty$.

Proof. Condition on the whole environment. Distinct apex labels use independent edge uniforms. There are $m^{\underline{k}}$ injective assignments of edge slots to the vertices of S. A union bound and the two-term Bonferroni inequality give

$$\mathbb{P}(Y_{S,t} = 1 \mid (\psi_j)) = m^{\underline{k}} \prod_{v\in S} p(v,t) + O_m\left(\max_{v\in S} p(v,t) \prod_{v\in S} p(v,t)\right). \tag{18}$$

Indeed, two distinct compatible assignments specify at least one extra edge; incompatible assignments have zero intersection probability. Since $t^x p(v,t) \to D_v$, the conditional mean of $N_S(T)$ is asymptotic to $m^{\underline{k}} \prod D_v\, T^\rho/\rho$. This follows by bounding the probabilities above and below by constant multiples of t^{-kx} after any fixed accuracy threshold and summing $\sum_{t\leq T} t^{-kx} \sim T^\rho/\rho$.

The centered variables $(Y_{S,t} - \mathbb{E}[Y_{S,t} \mid (\psi_j)])/t^\rho$ are independent for this fixed environment, and their variance sum is bounded, apart from a finite prefix, by $C(\psi)\sum_t t^{kx-2} < \infty$. The independent series criterion and Kronecker's lemma [2, Theorems 2.5.6 and 2.5.9] show that the centered partial sum is $o(T^\rho)$ almost surely. Fubini transfers this conclusion to the joint probability space.

For the second moment, two distinct apices $t, s > \max S$ joined to the same fixed set have probability at most $C_S t^{-kx} s^{-kx}$ by Lemma 3.1 with zero weight multiplicities. The diagonal probabilities are at most $C_S t^{-kx}$. Summing proves the uniform second-moment assertion. An almost-surely convergent family Z_T with bounded second moments converges in L^1: Fatou bounds the second moment of its limit Z, and Cauchy–Schwarz gives

$$\mathbb{E}|Z_T - Z| \le \varepsilon + (\mathbb{E}|Z_T - Z|^2)^{1/2} \mathbb{P}(|Z_T - Z| > \varepsilon)^{1/2}.$$

The probability tends to zero and then ε can decrease to zero. This proves (17). □

Let C_t count the induced octahedral k-sets in the graph on the distinct older neighbors of t, and let $C(T) = \sum_{3 \le t \le T} C_t$. These counts use only finite graphs, and $C(T) = \sum_S I_S N_S(T)$. For a cutoff K, let $C^K(T)$ retain bases $S \subset [K]$ and apices $t > K$, and let $A_K = \sum_{S \subset [K], |S|=k} I_S \prod_{v \in S} D_v$.

Proposition 3.5. *The complete count satisfies*

$$\frac{C(T)}{T^\rho} \xrightarrow{L^1} \frac{m^{\underline{k}}}{\rho} A. \tag{19}$$

Proof. At fixed K, Lemma 3.4, multiplied by each bounded I_S, proves $C^K(T)/T^\rho \to m^{\underline{k}} A_K/\rho$ in L^1. No independence between I_S and the weights is required. The required base edges and the k apex edges give

$$\mathbb{P}(I_S = 1, Y_{S,t} = 1) \le C t^{-kx} f_F(S),$$

summed over the finitely many matchings F on S. Therefore

$$\limsup_{T \to \infty} \mathbb{E} \frac{C(T) - C^K(T)}{T^\rho} \le C \sum_F \sum_{\max S > K} f_F(S). \tag{20}$$

The omitted apices $t \le K$ have count at most $K\binom{m}{k}$, which disappears before the displayed limsup. The right side tends to zero by Lemma 3.2, and $A_K \uparrow A$ in L^1. Taking $T \to \infty$ first and then $K \to \infty$ proves (19). □

4 Finite homology and link cores

For vertex t, let L_t be its link in X_t, the clique complex of the simple graph on its older neighbors. It has at most m vertices. The inclusions $L_t \subset X_{t-1}$ induce maps $f_{j,t}$ on H_j. The decomposition $X_t = X_{t-1} \cup \mathrm{St}_{X_t}(t)$ gives

$$\beta_q(t) - \beta_q(t-1) = \dim \ker f_{q-1,t} - \operatorname{rank} f_{q,t}. \tag{21}$$

This is the link-rank identity used in [4, Proposition 14 of the author preprint]. For completeness, the rational simplicial-chain sequence for a union of subcomplexes $\mathcal{K} = \mathcal{U} \cup \mathcal{V}$, $\mathcal{L} = \mathcal{U} \cap \mathcal{V}$, is

$$0 \longrightarrow C_*(\mathcal{L}) \xrightarrow{\zeta \mapsto (\zeta, -\zeta)} C_*(\mathcal{U}) \oplus C_*(\mathcal{V}) \xrightarrow{(\xi,\eta) \mapsto \xi + \eta} C_*(\mathcal{K}) \longrightarrow 0.$$

It is exact degree by degree because the simplex bases shared by $\mathcal{U}$ and $\mathcal{V}$ are exactly those of $\mathcal{L}$. If a cycle in $\mathcal{K}$ is lifted to (ξ, η), its boundary has the form $(\zeta, -\zeta)$, with ζ a $\mathcal{L}$-cycle. The class of ζ is unchanged by a different lift or a homologous cycle. This gives the connecting map; a zero connecting class is exactly a lift that can be modified into two cycles. The other two exact positions follow by lifting a bounding chain and subtracting it. Thus the sequence yields the homology exact sequence. A cone has zero reduced homology: adjoining its apex to

an augmented simplex gives a chain homotopy h with $dh + hd = \mathrm{id}$. Applying the sequence with $\mathcal{V}$ the star of t gives the exact segment

$$H_q(L_t) \xrightarrow{f_{q,t}} H_q(X_{t-1}) \longrightarrow H_q(X_t) \longrightarrow H_{q-1}(L_t) \xrightarrow{f_{q-1,t}} H_{q-1}(X_{t-1}).$$

Exactness at the middle terms gives

$$\dim H_q(X_t) = \beta_q(t-1) - \operatorname{rank} f_{q,t} + \dim \ker f_{q-1,t},$$

which is (21). The same calculation will keep track of the inclusion maps when vertices are deleted from a link.

4.1 The typical link

Put $d = k - 2 = 2q - 2$. Delete, repeatedly, vertices of a link graph whose current degree is less than d. The remaining graph is its d-core. Call L_t exceptional if this core has at least $k + 1$ vertices. Let $B(T)$ be the number of exceptional links up to time T. In homology expressions, an octahedral base S denotes its induced clique complex, rather than only its vertex set.

Lemma 4.1. *If L_t is not exceptional, then*

$$\dim H_{q-1}(L_t) = C_t \in \{0, 1\}, \qquad H_q(L_t) = 0.$$

When $C_t = 1$, its unique octahedral base S induces an isomorphism $H_{q-1}(S) \to H_{q-1}(L_t)$.

Proof. A flag complex with nonzero reduced H_j has at least $2j + 2$ vertices. For $j = 0$ this is disconnectedness. For $j \geq 1$, take an induced subcomplex minimal by vertices with nonzero H_j. Deleting any vertex makes H_j zero, so the homology exact sequence injects H_j into reduced H_{j-1} of its link. Induction gives at least $2j$ neighbors for each vertex. A universal vertex would make the complex a cone; hence there is also a nonneighbor, proving the lower bound $2j + 2$.

A vertex of degree below $2q - 2$ has a link on at most $2q - 3$ vertices. Its reduced H_{q-2} and H_{q-1} vanish by the support bound. The exact sequence therefore shows that its deletion induces an isomorphism on H_{q-1}. It also induces an isomorphism on H_q, by the same bound in the next degree. Every induced octahedral base survives the deletions, since all its internal degrees are $k - 2$.

If the remaining core has fewer than k vertices, both the relevant homology and its octahedral count vanish. With exactly k vertices, either there is a universal vertex, and the complex is a cone, or all degrees are $k - 2$. In the latter case its complement is a perfect matching, so it is the unique octahedral base. Its clique complex is a join of q two-point spaces: splitting at one missing pair expresses it as two cones over the $(q - 1)$-fold join. The reduced exact sequence shifts homology by one. Starting with two points shows that H_{q-1} is one-dimensional and $H_q = 0$. The isomorphisms under deletion give the asserted inclusion map. □

Lemma 4.2. *The exceptional links have $\mathbb{E}B(T) = o(T^\rho)$.*

Proof. An exceptional link contains a graph F on h vertices of minimum degree at least d, where $k + 1 \leq h \leq m$, and t is joined to all these vertices. There are finitely many ordered graph patterns for fixed m. The h apex edges add one incoming edge to each base vertex. Lemma 3.1, with zero weight multiplicities, therefore bounds the probability of a specified occurrence by

$$Ct^{-hx} \prod_{v \in F} v^{-(1-x)(1+d_-(v))-xd_+(v)}.$$

Lemma 3.2 sums the base labels. Summing the patterns gives $\mathbb{P}(L_t \text{ exceptional}) \leq Ct^{-(k+1)x}$. Thus $\mathbb{E}B(T)$ is $O(T^{1-(k+1)x})$, $O(\log T)$ or $O(1)$ according as $1 - (k + 1)x$ is positive, zero or negative. Each is $o(T^{1-kx})$. For $m = k$, no exceptional link is possible. □

4.2 Earlier cones and the Betti limit

Lemma 4.3. *Almost surely $A > 0$. For every K, each existing octahedral base $S \subset [K]$ has a common neighbor $s_S > K$ at a finite random time. The maximum of their finitely many first such times is finite almost surely.*

Proof. The occupancy calculation (18) applies to any fixed h-set with $h \le k$, and its conditional common-neighbor probability is asymptotic to a positive multiple of t^{-hx}. It has divergent sum because $hx < 1$. Conditional independence and the second Borel–Cantelli lemma [2, Theorem 2.3.7] give infinitely many common neighbors. If one fixed extra vertex is forbidden, the probability of also joining it is $O(t^{-(h+1)x})$, whenever $h+1 \le m$, so there are still infinitely many common neighbors avoiding it. These statements hold simultaneously for all fixed finite sets and extra vertices, by countability and Fubini.

A nonadjacent pair exists once a new vertex has more than m old vertices available. Starting from this two-point sphere, choose a common neighbor of the current octahedral sphere, then a second common neighbor avoiding the first. The two new vertices give its suspension as an induced subgraph. Repeat until there are k vertices. At each step the current sphere has at most $k-2$ vertices, so the required edges and the avoidance condition are within m. Edges between old vertices are never added later. This constructs an induced octahedral base. Its weights are positive, hence $A > 0$.

For a fixed K there are finitely many bases in $[K]$. Each has a common neighbor $s > K$ almost surely. Let τ_K be the largest of their first such labels, or K if no base exists. Then $\tau_K < \infty$ almost surely. The star of each such s contains the cone on its base, so the map from that base's homology into $H_{q-1}(X_{t-1})$ is zero for $t > \tau_K$. □

Proposition 4.4. *The limits in (7) hold.*

Proof. On a nonexceptional link after τ_K, if its unique base is in $[K]$, Lemma 4.1 identifies its entire H_{q-1} with that of the base. The latter maps to zero in the old complex by Lemma 4.3. Equation (21) therefore gives a birth of dimension one, and there is no death because $H_q(L_t) = 0$. All other nonexceptional links give nonnegative increments. On exceptional links both homology dimensions and base counts have a bound depending only on m, q, since the link has at most m vertices. Summing (21) gives, for a fixed constant M,

$$\beta_q(T) \le C(T) + MB(T), \tag{22}$$

$$\beta_q(T) \ge C^K(T) - C^K(\min(T, \tau_K)) - MB(T). \tag{23}$$

The prefix term divided by T^ρ tends to zero almost surely, because τ_K is finite. It tends to zero in L^1 as well: it is bounded by $C^K(T)/T^\rho$, and the latter has uniformly bounded second moments by Lemma 3.4 and the finite number of bases in $[K]$. The Cauchy–Schwarz estimate in that lemma applies. No expectation bound on τ_K is needed.

It follows from (22)–(23) that

$$|\beta_q(T) - C(T)| \le C(T) - C^K(T) + C^K(\min(T, \tau_K)) + MB(T).$$

Use (20) and Lemma 4.2, taking $T \to \infty$ at fixed K and then $K \to \infty$. This difference is $o(T^\rho)$ in L^1. Proposition 3.5 proves the first limit in (7); taking expectations proves the second. Its limiting constant is positive by Lemma 4.3. Dividing by that deterministic mean, which is positive for sufficiently large T, yields L^1 convergence of the mean-normalized variables to $A/\mathbb{E}A$ in this coupling, and hence the asserted convergence in distribution. □

5 Nondegeneracy and the upper tail

The coefficient retains random early structure. Its nonconstancy can be seen by making the weights of every vertex before a large fixed label small at the same time.

Proposition 5.1. *The coefficient A has essential infimum zero.*

Proof. Let $d = k - 2$ and $e = k(q-1)$, the degree and edge count of an octahedral base. Condition on $\psi_K = z$, for $K \geq 2$, and split the series into bases with a label below K, bases with minimum K, and bases entirely after K.

Recall that $\ell_K = r_K + d_-(K)$, while h_K counts weights below K and required edges crossing K in Lemma 3.1. For a single base, r_K is one if K belongs to it and zero otherwise. In the first group, at least one weight contains $1 - \psi_K$, so $h_K \geq 1$. Also $\ell_K \leq d + 1$. Summing its conditional bound using the same summable base weights (16) gives

$$\mathbb{E}[A_{\text{old}} \mid \psi_K = z] \leq C(1 + K^{d+1})(1 - z). \tag{24}$$

For a base with minimum exactly K, the vertex K has weight multiplicity one, incoming degree d and outgoing degree zero. The conditional cost is at most CK^{d+1}. The sum of all base exponents in (16) is $-(1-x)k - e$. Using (14),

$$\begin{aligned} \mathbb{E}[A_{\min=K} \mid \psi_K = z] &\leq CK^{d+1}K^{k-1-(1-x)k-e} \\ &= CK^{kx-e+d} = CK^{kx-2(q-1)^2}. \end{aligned} \tag{25}$$

Bases entirely after K, including their weights, are independent of ψ_K. Equation (13) gives

$$\mathbb{E}[A_{\min>K} \mid \psi_K = z] \leq CK^{kx-e}. \tag{26}$$

Both exponents in (25)–(26) are negative, since $kx < 1$, $e - d = 2(q-1)^2 \geq 2$ and $e \geq 4$. Given $\varepsilon > 0$, first choose K so that their sum is below $\varepsilon/2$, and then $\eta \in (0,1)$ so that the right side of (24) is below $\varepsilon/2$ when $1 - \eta < z < 1$. This Beta interval has positive probability. The conditional mean of A on the interval is below ε; Markov's inequality therefore gives $\mathbb{P}(A \leq 2\varepsilon) > 0$. This proves $\operatorname{ess\,inf} A = 0$. □

Together with $A > 0$ almost surely and $0 < \mathbb{E}A < \infty$, this proves that $W = A/\mathbb{E}A$ is nondegenerate. Proposition 3.3 gives every positive moment of W. For every $p > 0$ its survival function is $O_p(z^{-p})$, by Markov's inequality. To finish Theorem 2.2, suppose (9) held with a finite $\alpha > 0$. Its survival function must be positive eventually. For any $\epsilon > 0$, the ratio at $u = 2$ is at least $2^{-(\alpha+\epsilon)}$ for large z. Iteration gives a dyadic lower bound, and monotonicity extends it to $\mathbb{P}(W > y) \geq Cy^{-(\alpha+\epsilon)}$ for all sufficiently large y. An integer $p > \alpha + \epsilon$ then satisfies

$$\mathbb{E}W^p = p \int_0^\infty y^{p-1}\mathbb{P}(W > y)\, dy = \infty,$$

by nonnegative Fubini, contradicting Proposition 3.3. A bounded survival support also cannot satisfy (9). Thus no finite positive regular-variation exponent occurs in the strict regime of the theorem.

AI-use statement

OpenAI Codex was used extensively for literature review, development and checking of the arguments, and preparation of this manuscript. The mathematical claims are offered for independent review.

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